PARTIAL DYSON SPHERE

by

Patrick Brault

6/1/2025

A paper written for
general purpose on the topic of the massive structure known as the Dyson sphere. This, however, only discusses twenty-percent coverage of the Sun's surface at its poles.

I will show you how
it is possible to generate 42 Trillion kWh. In other words, 42 Quadrillion Watts of continuous power 365/24/7. Powering 34 Trillion Homes.

Showing, with little maintenance and a lot of setup, that it's all about equilibrium.



ABSTRACT

The sunlight that hits Earth, runs parallel to what is called the ecliptic plane. Sunlight that runs perpendicular to this plane is wasted. However, it can be captured, stored, transported and used anywhere in the solar system. Explaining how this all works together is the main purpose of this paper.

Picture the distance from the Earth to the Sun as the length of a football field, 100 yards. You can get to your opponent's 20-yard line without burning up. At this distance there are few materials that can withstand the 600 degree Celsius temperatures.

From wasted sunlight, at either of the Sun's poles, at a distance equal to the opponent's twenty-yard line, a square meter solar panel will produce 35000 watts. At Earth it is only 1354 watts per meter squared. From this simple change in distance to the Sun, I will show you how it is possible to generate 42 Trillion kWh. In other words, 42 Quadrillion watts of continuous power 365/24/7. Powering 34 Trillion homes.

The main components include a capacitor, a magnetron, an anode ring, orbital stations and a system of ultra-thin solar sheets. All components sit behind the solar sheet, which offers protection from the Sun. Beyond this, once completely setup, the only component that will need regular maintenance are the solar sheets. All other components require little maintenance.

Table of Contents

Chapter	Title	Pg
1	Current Distribution of Solar Energy	
	1.1 The Ecliptic Plane	0
	1.2 Solar Energy	0
2	Inefficiency of Current System	
	2.1 Atmospheric Interference. . .	0
	2.2 Solar Irradiance and Distance . .	0
3	Dangers of Current System	
	3.1 Wave Energy and Atom Ionization .	0
	3.2 Solar Radiation	0
	3.3 Solar Flares	0
	3.3.1 Communication Errors . .	0
4	Solar Irradiance Equation	
	4.1 Solar Irradiance Definition . .	0
	4.2 Solar Irradiance Equation . . .	0
	4.3 Solar Sheet Sample Size . . .	0
	4.4 Solar Sheet Full Size . . .	0
	4.5 Number of Homes Powered . .	0
5	Scalability	
	5.1 What is Scalable	0
	5.2 Component Parts	0
6	Mechanics	
	6.1 Solar Radiation	0
	6.2 Gravity	0
	6.3 Lightning Storm	0
	6.4 Capacitor	0
	6.4.1 Physical Parameters . .,	0
	6.5 Magnetron . . .	0
	6.5.1 Background . . .	0

6.5.2 Threshold Voltage . . 0
6.5.3 Efficiency . . . 0
6.6 Anode Ring 0
6.7 Free Space Path Loss . . . 0
6.8 Earth Systems 0
6.8.1 Transmitting Station . . 0
6.8.2 Receiving Station . . 0
6.8.3 Incoming Signal . . 0
6.8.4 Direct Current . . 0
7 Assembly and Material Logistics
7.1 Primary Transport . . . 0
7.2 Material Weight . . . 0
7.3 Launch Schedule . . . 0
8 Maintenance
8.1 General Maintenance . . . 0
8.2 Solar Sail 0
8.3 Magnetron 0
8.4 Capacitor 0
9 Efficiency
9.1 Overall Efficiency . . . 0
9.2 Component Efficiency . . . 0
9.3 Improvement and the Future . . 0
10 First Steps
10.1 The Venus Staging Area . . 0
10.2 Setting up 0
10.3 Moving into Place . . . 0
10.4 Building Heritage . . . 0
11 What About The Sun
11.1 The Sun's Health . . . 0
12 Theoretical Application of EM-Wave Technology
12.1 Anti-Gravity . . . 0
12.2 Phase Shifting . . . 0
12.3 Spontaneous Down Conversion . . 0

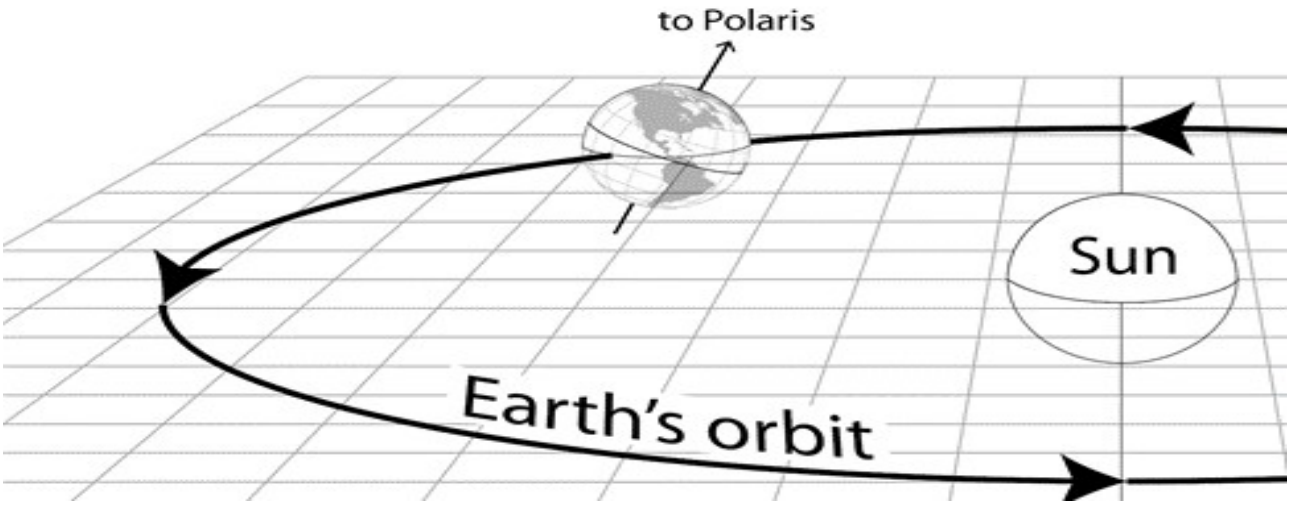
12.4 How this works 0
12.4 Applications 0

13 Variable List
13.1 List of Variables . . . 0
14 Definitions
14.1 List of Definitions . . . 0
15 Appendix
15.1 Appendix 0

Chapter 1: Current Distribution of Solar Energy

1.1 The Ecliptic Plane

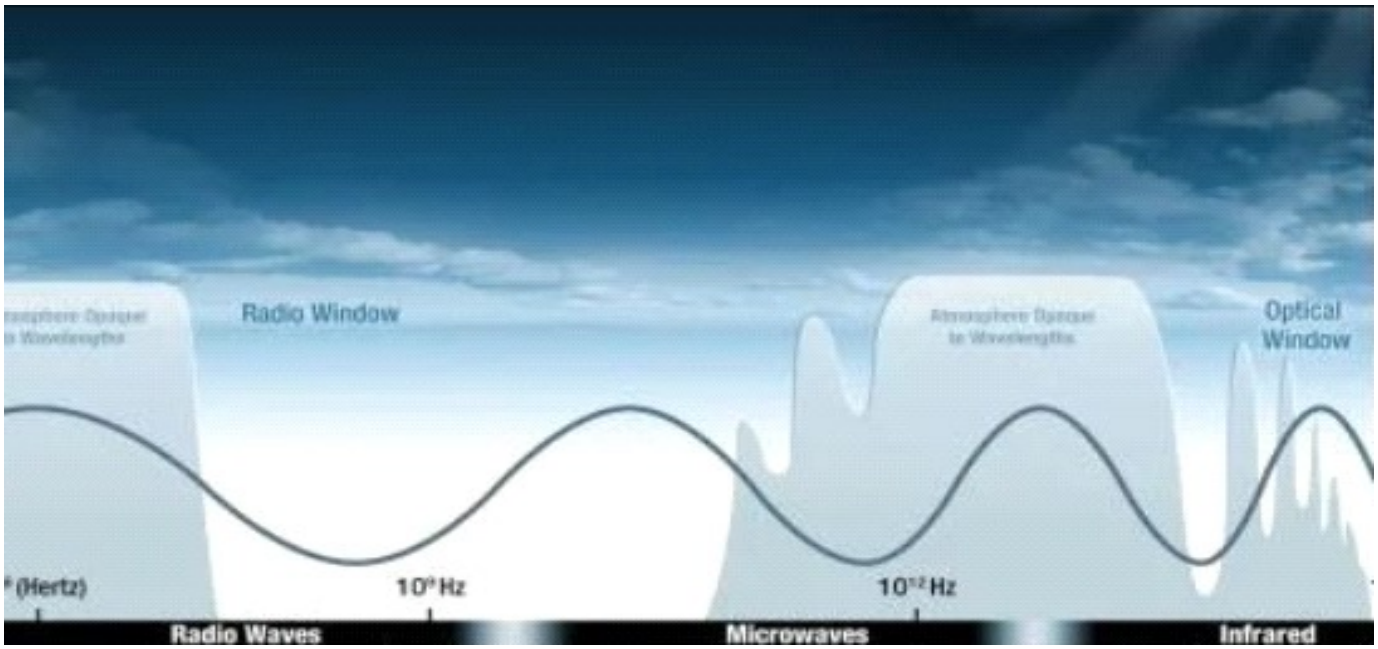
The ecliptic plane is the flat surface that the planet's orbits lay on as they rotate around the sun. All the planet's orbits sit on this plane, with few exceptions. This flat surface forms a grid, shown in the image below.



The planet's orbits sit on this grid, it is thought, because when our solar system formed, it formed from a flat rotating disk of gas and dust that was centered around the young Sun. As the dust settled, forming our planets, it did so in the overall shape of the dust cloud. Which our solar system is, just a spinning disk. This is important because this plane is where all our Sun's energy comes from. From the energy required to grow the grass on the ground, to the energy used by a solar panel to make electricity. It's all Solar Energy that comes from the Sun's equator, traveling along the ecliptic plane, until it reaches Earth.

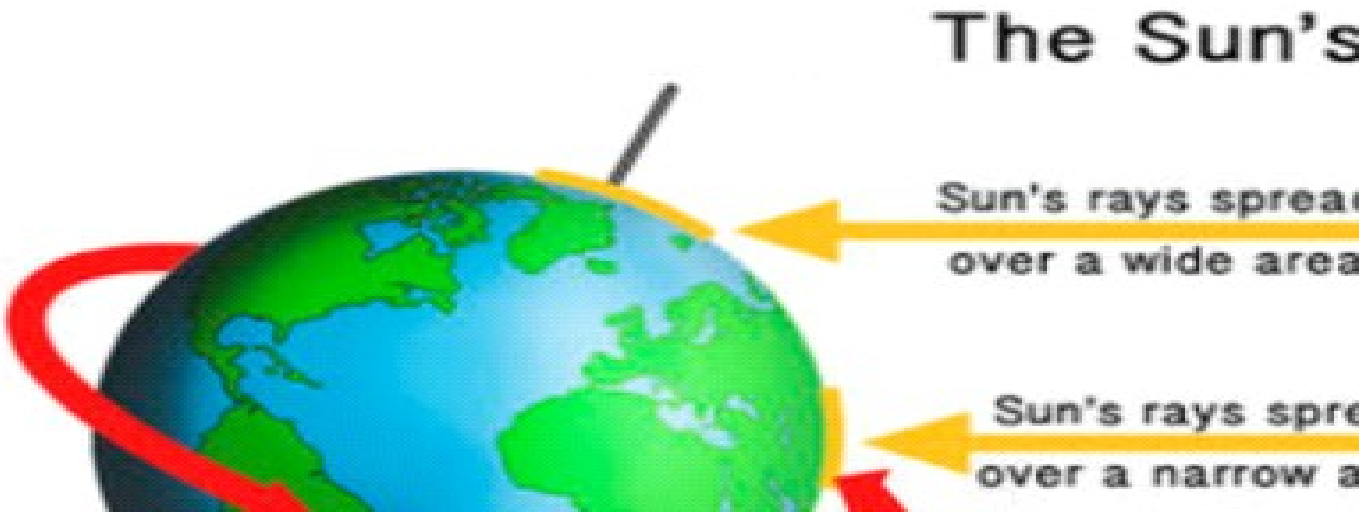
1.2 Solar Energy

Solar energy is measured in a few ways. Primarily, there is Electro-Magnetic Radiation, the stuff that is responsible for making auroras in our atmosphere. It flows through space as a wave of a specific frequency. For example, ultraviolet sunlight, the Sun's rays that cause sunburn, have a high frequency for an electromagnetic wave. In fact, a higher frequency than visible light, microwaves or radio waves. The UV frequency band sits just above the visible light frequency band, indicating that it has higher levels of energy. However, the next band of frequencies upwards from there, is X-rays, and it gets more dangerous from there. This hierarchy can be seen in the image below, note that Microwaves and Radio waves are at the low end of the spectrum. I will discuss dangers further in the section about wave energy and atom ionization.



The Sun radiates solar energy in every direction. That is, the Sun's surface radiates energy uniformly. Meaning the same amount of energy comes from every bit of the Sun's surface. It is not until later that all this energy, in the form of various frequencies of the electromagnetic spectrum, gets caught up in the interstellar wind and is carried away. The Sun's rays that come out of the northern and southern polar regions are identical to those on the equator.

Some of the Sun's rays make it to Earth, some don't. This paper is concerned with the ones that don't. These rays never hit a planet or get absorbed by anything. Instead, they are lost into space. Labeled as 'missing' in the image below.



The sunlight that hits Earth, runs parallel to, or along, the ecliptic plane. Sunlight that runs perpendicular to, or away from, the ecliptic plane is wasted. The latter is one type of inefficiency when speaking about collecting solar energy. No matter how many solar panels we put on or around Earth, the rays that run away from the ecliptic plane will always be lost. However, this lost solar energy can be captured, stored, transported and used anywhere in the solar system. Basically, solar panels collect sunlight, turning it into electricity. The electricity is then stored in a capacitor, which is connected to a magnetron. The magnetron creates an electromagnetic wave, or beam, targeting anode rings that direct the beam along the way to help focus and steer it toward a receiving station near Earth. This can be done twice, one beam created from Northern Solar Pole and another from the Southern Solar Pole. The beam runs above and below the ecliptic plane towards Earth, missing all the planetary bodies that lie on the ecliptic plane in between Earth and its Sun. An Earth based orbital station, one based atop the Earths Northern Pole and another below its Southern Pole will receive the EM-Wave, distributing it to terrestrial stations that connect to a power grid. All this and more will be explained in detail in a later chapter.

Chapter 2: Inefficiency of Current System

2.1 Atmospheric Interference

The Sun's missing rays are not the only inefficiency. Weather creates another type of inefficiency. Not only does the atmosphere itself scatter the Sun's rays. The weather, like clouds, scatters them even more. Weather can block sunlight completely and make Earth based photovoltaic systems non-effective. In the best cast, only about fifty percent of sunlight makes it to the surface, as can be seen in the image below.

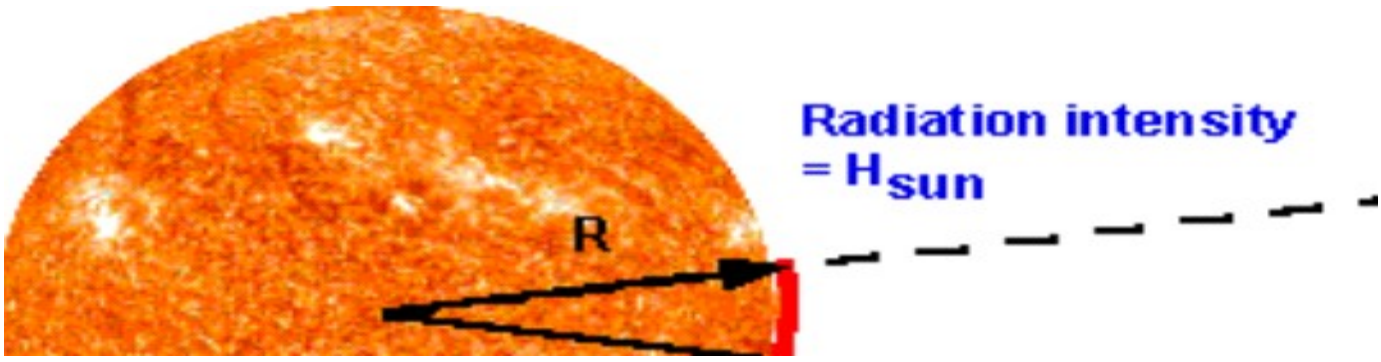


This type of interference can be thought of in terms of Peak Sun Hours. Where one peak hour indicates that a, one meter by one meter, solar panel will have 1000 Watts per Meter Squared (W/m^2) of solar radiation that is available to convert into direct current. This is important because of something called Solar Irradiance. Solar Irradiance is how much power the sunlight is carrying. It has been measured that, in the mainland United States, the average number of hours per day that are available in peak sun hours is only five hours. In other words, there are only five hours, out of every day, where the sunlight that reaches the surface will have a Solar Irradiance close to the value I am using for all my calculations. That being Solar Irradiance (Ie) = 1362 W/M^2, as shown in the table of variables and values. Furthermore, keep in mind that Solar Irradiance is zero at night, which I am not accounting for either.

Electro-Magnetic Waves, like radio waves, can pass through the atmosphere untouched. As can microwaves. They do not care about weather or other such interferences. So, if radio waves and microwaves are so closely related, why are a radio and a microwave so very different? I will address this, and other dangers, in a later section when discussing atom ionization. For now, just remember that radio and microwaves are closely related when viewed from the electromagnetic spectrum.

2.2 Radiation Intensity

As the Sun's rays spread out from its surface, they become less dense. Less concentrated with sunlight. This is based on distance to the Sun. In the image below, you can see how this plays out. The red line is small while close to the surface and then gets larger as distance gets further away. This is not only how sunlight spreads out but how it becomes less dense as it travels through space. It covers a larger area but the sunlight is less concentrated.



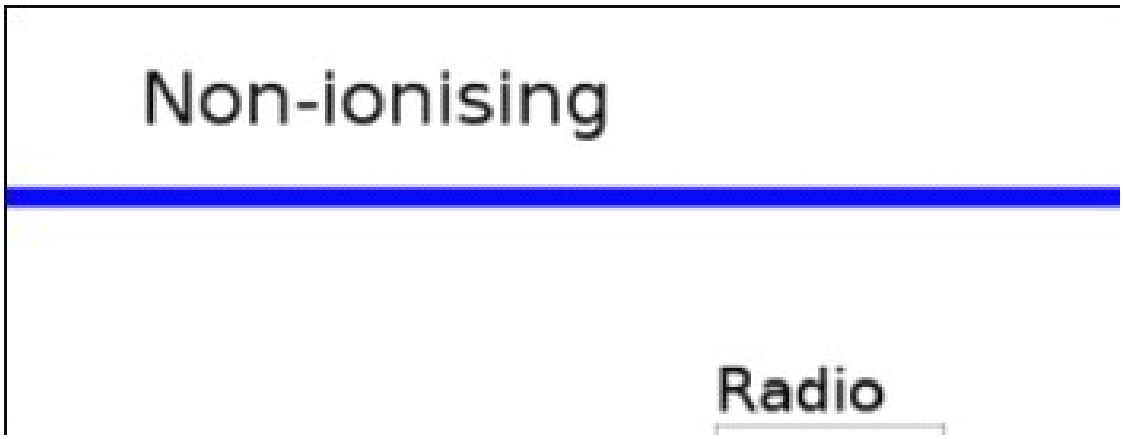
From this, it should be clear that the closer you are to the sun, the more power you will get from a solar panel. This is Solar Irradiance and will be discussed in detail in a later chapter.

At the smaller red line, on the Sun's surface, solar irradiance is equal to 6.32 x 10^7 watts per square meter (W/m^2). This is the starting value. The further away you get, the smaller that value gets. At Earth it is equal to 1362 W/m^2. In our picture above, assume D is equal to 29800000000 meters. This is the 20-yard line. It is as close to the sun as equipment can get without burning up. The high temperatures at this distance will be discussed later, however it can be said now that materials withstanding 600 degrees Celsius will be needed and some that come close are available. At this distance solar irradiance is equal to 34051 W/m^2. A big difference from the value at Earth.

Chapter 3: Dangers of Current System

3.1 Wave Energy and Atom Ionization

Based on how they interact with matter, electromagnetic waves can be broadly classified into two types; Ionizing and Non-Ionizing. Ionizing em-waves, like the x-ray and gamma ray, even a small amount, are harmful. The electromagnetic waves that are ionizing can be seen in the image below.



Both microwaves and radio waves can heat up a surface when they hit one. This is why your microwave oven heats up food. This will happen with a large microwave beamed from orbit to a ground station. I will discuss these stations and how the beam travels through our atmosphere in detail in later chapters. For now, concern yourself with the fact that stray microwaves are not much more energized than the radio waves floating around our atmosphere right now. Furthermore, none of these waves are ionizing. A stray x-ray could do molecular damage to a surface that it comes in contact with. As could a gamma ray. However, stray radio and microwaves will not.

3.2 Solar Radiation

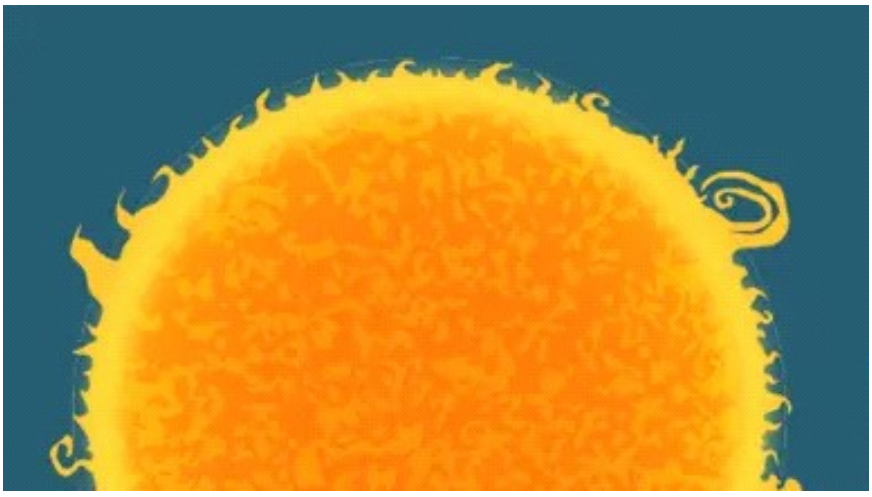
There are two main types of solar radiation. There is the kind that gives you a sunburn, this falls under visible and infrared light, and the kind that make the Northern Lights, which falls under radio and microwaves. Ionizing waves rarely make it to Earth.

A sunburn is short-term overexposure, while premature aging and skin cancer are side effects of prolonged exposure to solar radiation. As opposed to electromagnetic radiation, which causes dancing walls of lights in the sky. Both non-ionizing. The distribution of the different types of radiation, displayed in the image below, shows that most of the solar radiation is visible and infrared light.



Electromagnetic radiation, by comparison, is less common.

Cosmic radiation is electromagnetic radiation. It is emitted during solar events, like sunspots, solar flares, coronal mass ejections and proton storms. Whereas, visible and infrared light are emitted by the nuclear reaction inside the Sun. These are examples of extreme Sun activity that can send bursts of energy toward Earth.



The above image shows that this radiation moves in one direction in a straight line, along the ecliptic plane. Capping the solar poles with ultra-thin solar sheets has no effect on this radiation. Except that, at the 20-yard line from a solar pole, is above the ecliptic plane. Therefore, Em-waves that move from this point to Earth, traveling in a straight line, will have a path free of other planetary bodies. Not shown in the image above.

3.3 Solar Flares

Solar flares are a special kind of interference. They are a sudden and intense burst of electromagnetic radiation from the Sun's atmosphere and produce high energy particles and radiation that create dangerous geomagnetic storms, radiation exposure, and disrupt equipment. These intense bursts of energy occur more towards the equator than the poles. A focused burst that, if it makes it to Earth, will harm ground based electronic systems. However, the Earth's magnetic field, generated by our spinning molten iron core, acts like a defensive shield. This action can be seen in the image below, in how the solar winds push the magnetic field lines. Blocking harmful rays and diverting their paths around Earth. Again, capping the solar poles has no effect on solar flares that run along the ecliptic plane. However, as discussed in the section on Em-Technology and the Future, this defensive grid can be mimicked using the technology and scope of this paper.



Severe damage to our electric grid has occurred. Luckily, intense activity on the surface of the Sun is predictable and follows a repeating pattern. With this information it should be easy to see that this is a serious problem but one that we can do something about.

3.3.1 Communication Corruption

Minor damage from solar flares are more common. Solar flare radiation can disrupt radio signals and GPS communications. It causes data corruption in our communication systems. To understand this, let’s look

at an example. Consider a data packet traveling from your internet service provider to your home internet router. It is going to get corrupted. One-way corruption happens is simply the amount of wire that the signal has to go through to get to your computer. As the packet travels, it encounters interference. This interference changes the signal. It changes the data packet. So that now, your computer doesn't really know what it is. I am explaining simple data corruption. For this example, what we do to fix this, is send ten packets instead of one. This way, out of the ten packets that are sent, your computer will compare them and be able to figure out what exactly is trying to be sent. Same thing happens with stray particles. Stray x-rays and gamma rays are rare but very dangerous in this way too. Microwaves and radio waves are everywhere but not dangerous but can disrupt sensitive electronics. One solution to this is to wrap a cable in an aluminum foil wrapping. So, data corruption from stray waves is a problem already. If we could 'sift' out the bad stuff, then we could send less packets and not have to wrap electronics in aluminum foil. Which would increase the speed of our communication. Which means faster internet for everyone. Which also means an atmosphere full of less dangerous wave frequencies.

To wrap up this example, just remember that electromagnetic waves come in a spectrum of frequencies. Some are ionizing and some are not. Earth's magnetic field protects us from the ionizing stuff but if a big enough solar flare happens, then bad stuff gets through our natural defenses and can disable sensitive equipment.

Now to discuss solar proton events. They are a sudden increase in energetic protons. These are particles with energies greater than 10 MeV that reach Earth. Technically, this is a third type of radiation that comes from the Sun but happens with solar flares. These are subatomic particles, not electromagnetic waves. Storms of this type are very rare but do happen and will destroy equipment.

Chapter 4: Solar Irradiance Equation and Calculations

4.1 Solar Irradiance

Solar Irradiance is needed to calculate how much power a solar panel creates. Solar irradiance refers to the power per unit area received from the Sun's rays. Again, the closer you are to the sun, the bigger this number is. The main idea is that the closer you get to the Sun; the more wattage you can get out of the same equipment.

4.2 Solar Irradiance Equation

I'm going to use the short version of this equation. It uses Average Solar Power (Savg) as an input in place of some more complicated numbers. Answer comes out the same. So I'm using the easy version.

Solar Irradiance (I) = Savg / (4*pi*D^2)

Distance (De) to Earth = 15E^10 Meters

Distance (Dy) to 20 yd line = 2.9E^10 Meters

Average Solar Power (Savg) = 3.8E^26 Watts

4.3 Solar Sheet Sample

Picture a bunch of ultra-thin solar cells, connected together to form a large sheet. A high temp resistant fabric can be used as a backing where needed. This sheet is a circuit. It is a power source. A battery. The solar cells, in series and/or parallel connect together to become on large direct current power source. Weight costs will be discussed in a later chapter.

A solar sheet, in this fashion, one that is a hundred Kilometers by a hundred Kilometers has the following area.

Area (As) Sample = 1E^10 Kilometers Squared

That's an area of 10000000000, or ten billion, Meters^2. Which means ten billion small solar sails can fit in this area. It's a square shaped sheet. In real life, at full scale, the sheet well need to be an overall circular shape. I will discuss the full size sheet in the next section. The radius, of the sample sized sheet, if it where circular, would be about 50 km. To put this size in perspective, the image below shows ten of the most populated cities around the world, with a green circle marking the 50 km radius.


1. The Pearl River Delta, China
63.9M
2. Dhaka, Banglad
60.3M
Bangladesh
Dhaka
4. Shanghai/The Yangtze River Delta, China
58.4M
5. Kolkata, India
54.2M
5. Cairo/Alexandria/The Nile River Delta,

Next, to calculate Solar Irradiances for the distance to Earth and to the 20 yd. line. Remember, at Earth, Solar Irradiance comes out to:

Solar Irradiance (I) = 3.8E^26 Watts / [4*pi*(1.5E^11 Meters)^2] = 1344 W/m^2.

So, at Earth our solar sail sample size makes,

1344 W/m^2 * 1E^10 m^2 = 1.3E^13 Watts

Remember, at the 20-yard line:

Solar Irradiance (I) = 3.8E^26 Watts / [4*pi*(2.9E^10 Meters)^2] = 35000 W/m^2

So, at the 20 yd. line our sample sail makes,

35000 W/m^2 * 1E^10 m^2 = 35E^13 Watts

A big difference, if you ask me. Next, let's see what this looks like at full scale.

4.4 Solar Sheet Full Size

Now, let's see what happens with the fully proposed size, ten percent coverage of the Sun's surface at each pole. For a total of twenty percent coverage of the Sun's surface. The total area would be:

Area (Af) Full = 1.2E^12 Kilometers Squared

That's a total area of 1.2E^12 Meters Squared area of solar panels. Remember, this is both the Sun's polar region areas. So at the 20 yd. line, at full scale, the sails make:

35000 W/m^2 * 1.2E^12 m^2 = 4.15E^16 Watts

That is 42 Quadrillion watts of continuous power 365/24/7. It is equivalent to 42 Trillion kWh. Ok ... this should really blow your mind. That is a lot of electricity. Next, I will put this number in terms of how many homes we can power.

4.5 Number of Homes Powered

I will be using an average power usage per household in the USA of 1224 watts per home. Remember, at Earth, our sample sized solar sail makes 1.3E^13 Watts. So for a sample size solar sheet based in Earth orbit, the Number of Homes Powered (Hes):

Hes = 1.3E^13 Watts / 1224 Watts per Home = 11 billion homes

At 20-yard line, for the 100km x 100km sample sail. Remember, at the 20 yd. line our sample sized solar sail makes 35E^13 Watts. So, for our sample sized solar sheet, based at the twenty-yard line, th Number of Homes Powered (Hys):

Hys = 35E^13 Watts / 1224 Watts per Home = 290 billion homes

At full scale, remember, at full scale our solar sail makes 4.15E^16 Watts. The Number of Homes Powered (Hyf)

Hyf = 4.15E^16 Watts / 1224 Watts per Home = 34 trillion homes

As you can see, at full scale and with technology that is current, an additional 34 trillion homes could be powered.

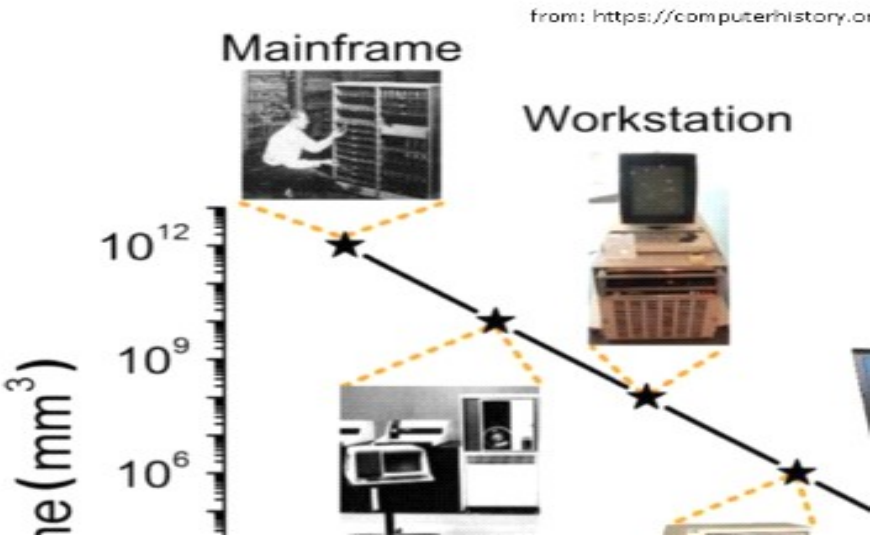
Chapter 5: Scalability

5.1 What does Scalable Mean

Scalability is the capacity of a system to manage increasing loads or demands without sacrificing performance. All of our equipment meets these criteria and is considered scalable because their main operations are based on fundamental physical laws that generally apply across different sizes and scales. This allows engineers to use knowledge gained from small-scale experiments to predict and design systems at much larger scales. Much like I am doing with the different sized solar sheets and why I can make a big space capacitor on paper.

5.2 What is Scalable

Component parts, like the, magnetron, capacitor, solar sail and lightning rod are scalable. I will be discussing these parts in detail in the next chapter. For now, just know that they can be made as large as we can possibly make them. The opposite has been done with computers. They have been made smaller and smaller, as can be seen in the image below.



This is possible, not only because our manufacturing abilities are able to make things smaller and smaller, but because science tells us that it will work. For example, it doesn't matter what you use as a switch, to compute on, or its size. Whether it be a vacuum tube, very large, or silicon, very small. It does the same thing.

Chapter 6: Mechanics

6.1 Solar Radiation

This section describes how much force the Sun pushes on the solar sail. This is important because this will help keep the entire solar sail in place vertically. Keeping it in one place will be kind of like using a LaGrange point between planetary gravitation. However, instead of using only gravity, the push of the Sun's solar radiation can also be used as a holding force.

The pull force of the Sun's Gravity will be calculated next. If these two values equal each other, then the entire solar sail will be held in place vertically. This push and pull effect can be visualized in how the Earth's gravity pulls you down but how a parachute keeps you up. See the image below.



There is one more mechanism to keep the entire sail structure floating above a Solar Pole. That is the use of a high specific impulse engine. That is, an engine with low but constant thrust. A prototype that has recently been developed, called Ikaros, will do this. Ikaros is an acronym for Interplanetary Kite-craft Accelerated by Radiation of the Sun and will keep things in place horizontally. Ikaros will provide stabilizing in the forward, backward, left and right direction. Remember that the Sun's solar radiation and gravity does this in an up and down direction. See image below of how Ikaros might look flying around Earth.



https://www.space.com/25800-ikaros-solar-sail.html is a link to some info on Ikaros.

Now to begin our calculations to find out how much force comes from solar radiation, the force pushing away from the Sun. I am going to skip the calculations for the sample sized sale and jump right into the full scale size.

Before I get to the big calculation, there are some parts we need to understand. Solar Radiation Intensity or Solar Irradiance Intensity (I) was described earlier. That a solar panel produces more energy the closer it is to the Sun. Remember, Solar Radiation at the 20 yd line it is:

Iy = 35000 W/m^2

The next number we need is the speed of light (c).

c = 3E^8 Meters per Second.

One more number, a new number, then we will get to the force on our solar sail. Instantaneous Pressure (p) is the pressure exerted by a force that is acting on a surface. In our case, that means it is the pressure on our solar sail exerted by the intensity of the Sun's radiation. It's applied over the whole area as a gentle push away from the Sun.

Instantaneous Pressure (p) = Solar Irradiance (Iy) / Spd of Light (c)

p = 35000 Watts per Meter Squared / 3E^8 Meters per Second = 1.13E^-4 Newtons per Meter Squared

Finally, to calculate the number at hand. The force on our full size solar sail from the intensity of the Sun (Fi). The force is calculated by multiplying the Instantaneous Pressure (p) with the area of our sample sail. You can track the units: [Newtons divided by Meters Squared] multiplied by [Meters Squared], after canceling things out this leaves only Newtons.

Fi = Instantaneous Pressure (p) * Area (Af) = 1.13E^-4 N * (1.2E^12 m^2)) = 1.4E^8 Newtons

This is the force (Fi) on solar sheets being pushed away from the sun.

As you will see from the next section, the pull from gravity is 2.7E^10 Newtons. That is a difference of 2.75E^10 Newtons. There is more force pull on our sail than pushing. This is important because this is where we get to play with weight and surface area costs. Weight is going to change; it will get smaller. Weight from the capacitors and magnetrons and all the other equipment will add to the weight costs. To offset this, we can increase surface area, which will increase the push from the solar radiation. Most improvements, to weight and efficiency, will come with time. First steps will be discussed in a later chapter. Building system heritage, however, is the fine tuning stage and is done prior to implementing a full scale project. This building heritage stage will be described later in detail in the same chapter.

6.2 Gravity

The amount of gravity that the Sun pulls on the sail needs to be calculated because the Sun's Gravity is one of the forces that will help keep the whole sail in place. Like in the image below, gravity pulls down, while solar radiation plays the role of the ground and pushes up.



The calculation is pretty simple. It only involves the masses and distance between the solar sail and the Sun. The mass of the Sun (M1) is generally accepted as:

M1 = 2E^30 Kilograms

The mass of the sample solar sail (M2a) is based on a panel that is available on the market right now. The market panel has an area (Ss) of 1 m^2 and a weight (Sm) of 2 kilograms. Based on this, the estimated weight of a sample solar sail built out of this heavy market panel is:

M2a = 2E^10 Kilograms

The mass of the proposed full scale sail (M2b), also based on the heavier market panel is:

M2b = 2.4E^12 Kilograms

The mass of the full scale (M2c), based on non-manufactured technologically available solar cell is:

M2c = 1.8E^11 Kilograms

The distance (Dy) between the sail and Sun, the one that I will use, is:

Dy = 2.9E^10 Meters

The radius of the Sun (Rs) is:

Rs = 7E^8 Meters

The Radius of the Sun should to be added to the number I am using. The number I am using is the distance to the Sun's surface. Which is where the radiation comes from. Right now though, I need to use distance to the center of the Sun, which is where gravity comes from. However, this is such a small difference, at this point, it can be ignored. The final number is the gravitational constant (G). Which, because gravity is constant everywhere, is just a number that can be found by experiment.

G = 6.7E^-11 ((Newtons * Meters^2) / kilograms^2)

Now comes the big equation where we use all these numbers, with the gravity equation. So, the Sun pulls on our sample size sail with this much force (F2a):

F2a = G * ((M1 * M2) / De^2) = 6.7E^-11 * (2E^30 * 2E^10) / (2.9E^10)^2) = 3E^9 Newtons

So, the sun will pull the sample solar sheet down with 3E^9 Newtons of force.

Next, to find the force on the full scale sail, also based on the market panel (F2b). The only number that is different in this calculation is the mass of our solar sail (M2b). To find this we need a few numbers:

Full Sheet Area (Af) = 1.2E^12 m^2

Area of the Market Available Panel (Ss) = 1m^2

Weight of the Market Available Panel (Sm) = 2 kg.

This then gives the weight of our full size solar sail built with market available panels to be:

M2b = Af / Sa * Sm = 1.2E^12 m^2 / 1 m^2 * 2 kg = 2.4E^12 kg

Then, the force that the Sun's gravity pulls on the full scale market based solar sail (F2b):

F2b = G * ((M1 * M2b) / De^2) = 6.7E^-11 * (2E^30 * 2.4E^12) / (2.9E^10)^2) = 3.7E^11 Newtons

Next, to find the force on the full scale sail, but now based on a technologically available solar panel that is not available on the market. Again, the only number that is different is the mass of our high technology solar sail (M2c). To find this we need a few numbers:

Full Sheet Area (Af) = 1.2E^12 m^2

Area of the Market Available Panel (Sss) = 1 m^2

Weight of the Market Available Panel (Sm) = .15 kg.

This then gives the weight of our high tech full size solar sail, that is floating over one pole, to be:

M2c = Af / Sss * Sm = 1.2E^12 m^2 / 1 m^2 * .15 kg = 1.8E^11 kg

Then, the force that the Sun's gravity pulls on it (F2c):

F2c = G * ((M1 * M2c) / De^2) = 6.7E^-11 * (2E^30 * 1.8E^11) / (2.9E^10)^2) = 2.7E^10 Newtons

Since our last number is technically possible, it is ok for me to use that number from here forward. All the numbers before it, are to help you understand how we get to that final number, in steps, or stages.

The technologically available solar cell could be placed on a backing of composite fabric, one that weighs only 13 grams per square meter, commercially known as Dyneema. If exposed to temperatures above 350°C (662°F), Dyneema can burn. Which almost gets us to the 20 yd. line, so there is a little work to be done here, we have a little further to go. However, Dyneema is a great start. That is just the wrapper for the actual solar cell though.

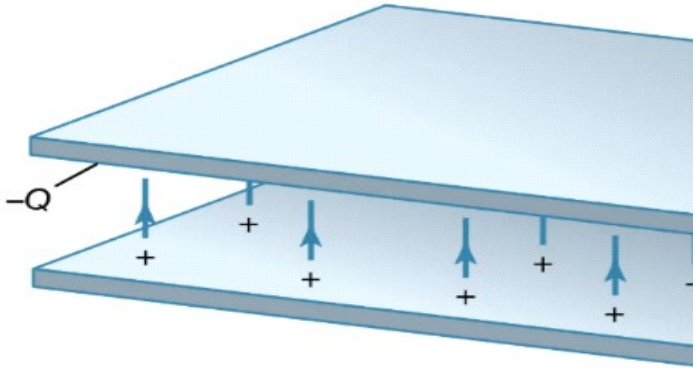
MIT researchers found the cell could generate 730 watts of power per kilogram when freestanding and about 370 watts-per-kilogram if deployed on the high-strength Dyneema fabric, which is about 18 times more power-per-kilogram than conventional solar cells. Generally, typical solar panels weigh around 10 kg to 20 kg or 22 to 44 pounds per square meter. The thin-film solar cells weigh about 100 times less than conventional solar cells, weighing in at .15kg per square meter. Here is a link to the paper.

6.3 Lighting Storm

Either the solar sheet is hard wired to the capacitor. Or, the massive amount of voltage we have, traverses the short distance to our capacitor, as a lightning storm. Between two lightning rods, one as the anode and the other as the cathode. Remember, all this equipment is safely protected behind our solar sheet. The solar sheet is what is taking most of the forces from the sun. Sitting behind this 'shield' is our capacitor and magnetron and all other equipment. It is possible to simply hardwire the solar sheet to the capacitor and then to the magnetron. However, let’s explore what a big space storm of electricity would look like.

6.4 Capacitor

A capacitor is a critical component in this, very large, electrical circuit. It has the ability to store energy in the form of an electrical charge, producing a potential difference, a static voltage, across its two metal plates, shown below.



The two metal places are parallel and separated by a dielectric, like air. Except in this case the dielectric will be empty space. When an external power source, like a solar cell, is connected to a capacitor it begins to charge the capacitor. When a capacitor is completely charged, a potential difference exists between its plates. One positive, the other negative. This is where electrons are stored.

6.4.1 Physical Parameters

Capacitors with different physical parameters can hold different amounts of charge when the same amount of voltage is applied across the capacitors. This ability of the capacitor is called capacitance. Capacitance (C) implies an associated storage of electrical energy. The capacitance is the ratio of the amount of Charge (Q) on either conductor to the Potential Difference (V) between the conductors, or simply C = Q/V. I will discuss potential difference shortly. To keep the positives and negatives in order, it should be noted that, the terminal of the capacitor that is connected to the cathode of the battery will get positively charged (+Q) and the terminal that is connected to the anode of the battery will get negatively charged (-Q).

Another term that will be needed to describe how all this energy is stored is the Potential Difference. It is the voltage between the two plates, their difference in electric charge. It will be equal to whatever power is coming from the solar sails, which is calculated below. More importantly, we can change this value by changing parameters, like distance between plates.

Because the charge (Q) coming from the solar sails is, more or less, equal and constant, the voltage drop across the capacitor is dependent on the Capacitance value in the equation, V = Q/C. Moreover, it is independent of Q or V. Which means, a lower capacitance value results in a bigger voltage drop, whereas a large capacitance value results in a smaller voltage drop. This is how the storage of the electricity is represented mathematically and is how we know which parameters will change and which won't.

Another important parameter of the capacitor is the area of the plates. The larger the area of the plates, the greater the charge that the capacitor can carry and the greater its capacitance. This can also be accomplished by making the distance between them smaller.

A capacitor is a simple battery. But one that can be made as big as you can imagine. It's true. We could make a capacitor the size of the USA. A capacitor of such size could be helpful in storing 42 Quadrillion watts. The capacitor is important because it holds the electricity in a place where we can use it. It also allows for a controlled way to steadily deliver power to other components like the Magnetron.

The coupling of a non-rigid structure made of copper and small electrical 'storms' going from anode to cathode is one way the electricity could flow from solar sails to capacitor. Adding resistors to control the flow of current, makes this an R-C-R circuit. A resistor - capacitor - resistor circuit. Because energy coming from the Sun is so volatile, something like this will be required.

To calculate Charge on Capacitor (Qc), we need our 100km x 100km solar sail as an example. I will also need to make some assumptions about the properties of our capacitor. Am assuming a 1000-meter distance between capacitor plates and space as what's between them. So, this give a Capacitance of 8.854E^-12 F/m * (100000 m / 1000 m) = 8.9e^-10 Farads. Using Capacitance, now we can calculate the Charge on the Capacitor (Qc). To make this calculation, I am assuming that R = %10.

Qc = 8.9E^-10 F * 6E^6*(1 - e^(-inf / (6E^5*8.9E^-10))) = .005 Coulombs

Let it be noted that the capacitor and magnetron are hard wired together. By good old copper wire where necessary.

6.5 Magnetron

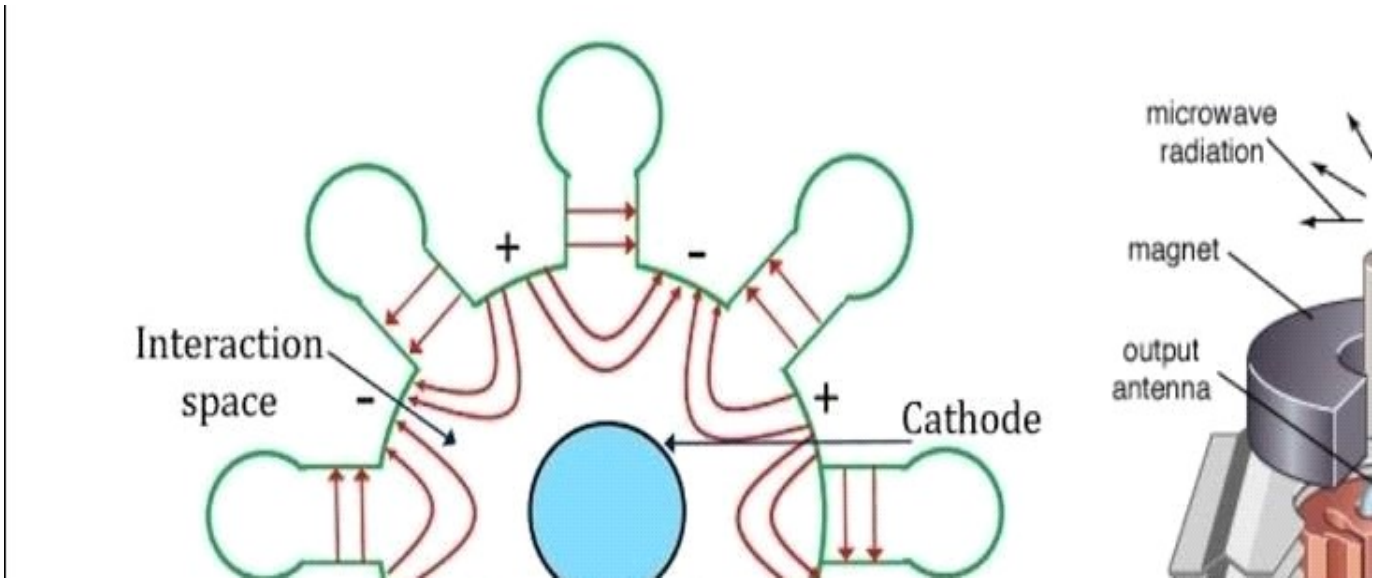
We will be using a Cavity Magnetron to transport energy through space to Earth. The Magnetron is part of a group of equipment known as the Electron Tube Family. They are tools for an under explored branch of physics, known as electricity and magnetism. Optics is the other unexplored branch of physics. Basically, you can think of this equipment as an electro-magnetic wave gun. I am following the work of Albert Hull in his book The Magnetron, published in the Journal of the American Institute of Electrical Engineers in September 1921, and H. Boot and J. Randall in their book The Cavity Magnetron, published in July 1946. All of the equations that I will be using in this section come from the second.

6.5.1 Background

First, some terms. The main function of the cavity Magnetron is that it takes advantage of valence electrons in a charged substance. Resonating chambers created in this substance govern the possible physical

paths for electrons by using rf waves and magnetic field lines. These paths flow over resonating chambers, sort of like how air flows over the inner workings of a whistle.

Other important terms that will be used are displayed in the image below; Interaction Space, Electric Field, Radio Frequency Field (RF), Path of Electron, Output Antennae, Magnetic Field (Magnet), Anode, Cathode, Teeth, Cavity. Every one of these represents a value for a specific aspect of the Magnetron or the operation of it. These terms and numbers will be referred to as the configuration of the Magnetron. In the above mentioned texts, configuration of the magnetron is coined using the term Mode. I am introducing a new term, Configuration, here because Mode is not completely explained.



6.5.2 Threshold Voltage

So how does our 42 Quadrillion Watts get from the Sun to the Earth!? For this, all we need are some starting values, we will need to make some assumptions, which I will explain. But first, I will need some equations and will be taking these from the text by Boot and Randall. They have shown that efficiency is highly dependent on the configuration, or mode, of the Magnetron. Therefore, I will need to discuss efficiency too.

Equations govern the operation of our Magnetron, they explain it. Threshold Voltage (Vt) is an important term. It is the level of energy that an electron has to reach in order to escape the Magnetron chamber, leaving through the barrel. This is based on the various dimensions of the magnetrons working components. Remember, efficiency is highly dependent on Configuration, or Config (or Mode), so I will describe the workings of the inner chamber in a later section. For now, the following equation will be used to determine Threshold Voltage.

$Vt * \lambda^2 / (10100 * a^2) = 2 * H * \lambda / n * (1 - (b^2 / a^2))/21400 - 1 / n^2 = e * Vt / (2 * pi^2 * m * c * a^2$

Where, a is anode radius, b is cathode radius, H is magnetic field in Oersteds, n is the number of cavities, λ is wavelength in Centimeters. To determine Anode Voltage (Va), we will be using:

Va = 4 * pi^2 * c^2 * m / e * (D^2 / n^2 * λ^2 * Φ^2)

Where, D is anode hole diameter in Centimeters, Φ is phase (or angle) between adjacent cavities.

Everything will be assumed to have been running a long time already. This is what is meant by time (t) = infinity. These are some very complicated equations, so for now, we will just look at them. However, after making some assumptions on the physical dimensions of the inner workings of our magnetron, I will be able to put them to some use.

Remember, that we are viewing the entire system at t = infinity. At this point, the current is zero (except for whatever the Magnetron is pulling), voltage is stable, all RF and EMF Fields are aligned, capacitor is charged, ... everything is in a state of equilibrium. At this point, we can begin to discuss the efficiency of the magnetron.

6.5.3 Efficiency

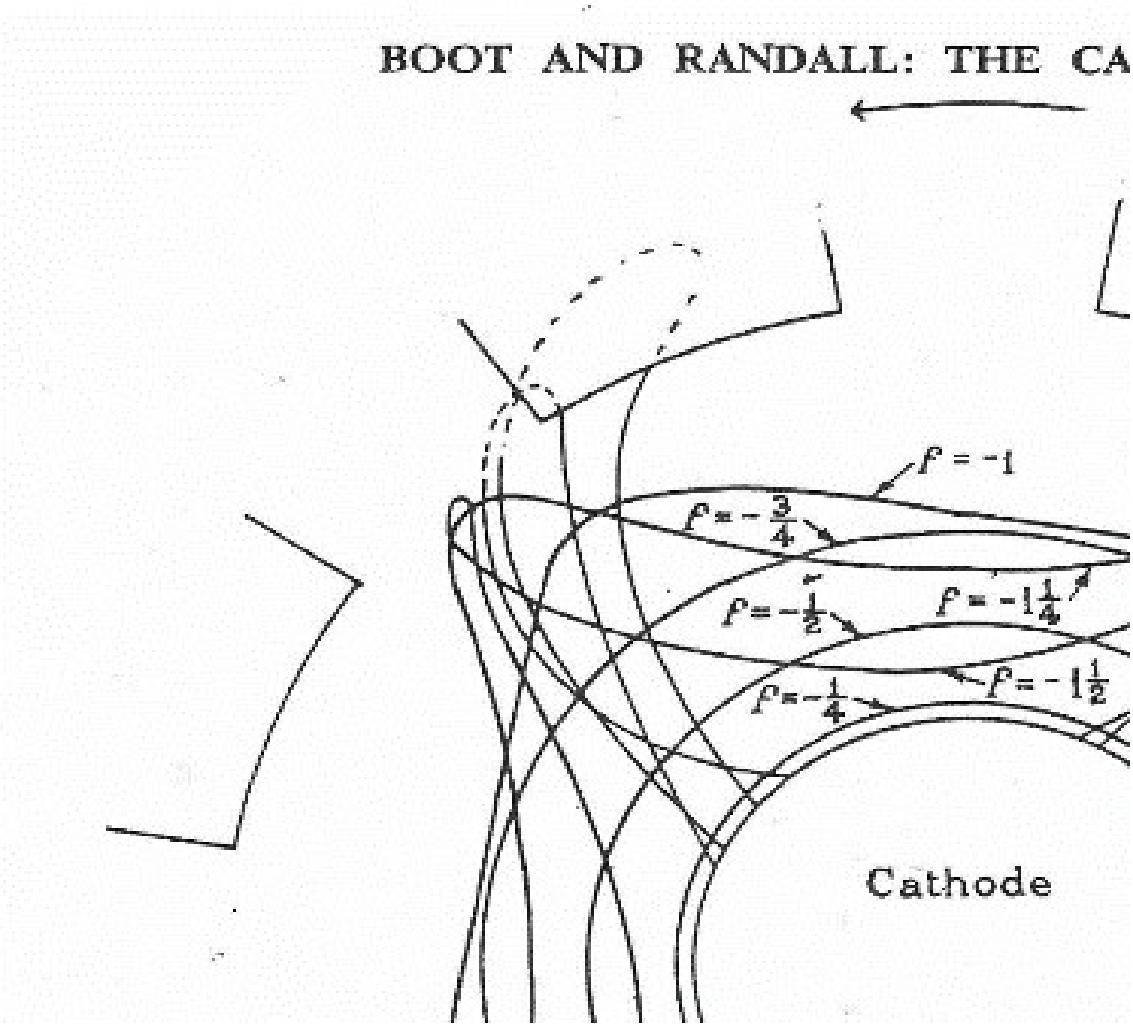
Efficiency of the magnetron comes down to one thing, how fast it can spit out electrons. Both images below, outline the path an electron takes when inside the magnetron. First, a drawing by Boots and Randall that shows the path an electron takes in order to get up to the point where it can break the voltage barrier, the Threshold Voltage. Again, this is where an electron has enough energy to escape the inner magnetron chamber through a barrel opening. The second, an image I made in the same fashion as Boots and Randall. However, I go one step further and attempt to explain possible improvements in the resonance chamber. Boots and Randall discuss magnetic field strength, they introduced coupling and linking the teeth, as an important step to increasing efficiency, and the number of chambers. They do not discuss chamber shape or the resonating em-wave inside it as a possible increase.

Increasing the efficiency of the magnetron means increasing the efficiency at which our electron is pushed around. Which means the RF field is a big player in this game and should not be ignored. I propose that the resonance chamber is an additional component where efficiency can be increased.

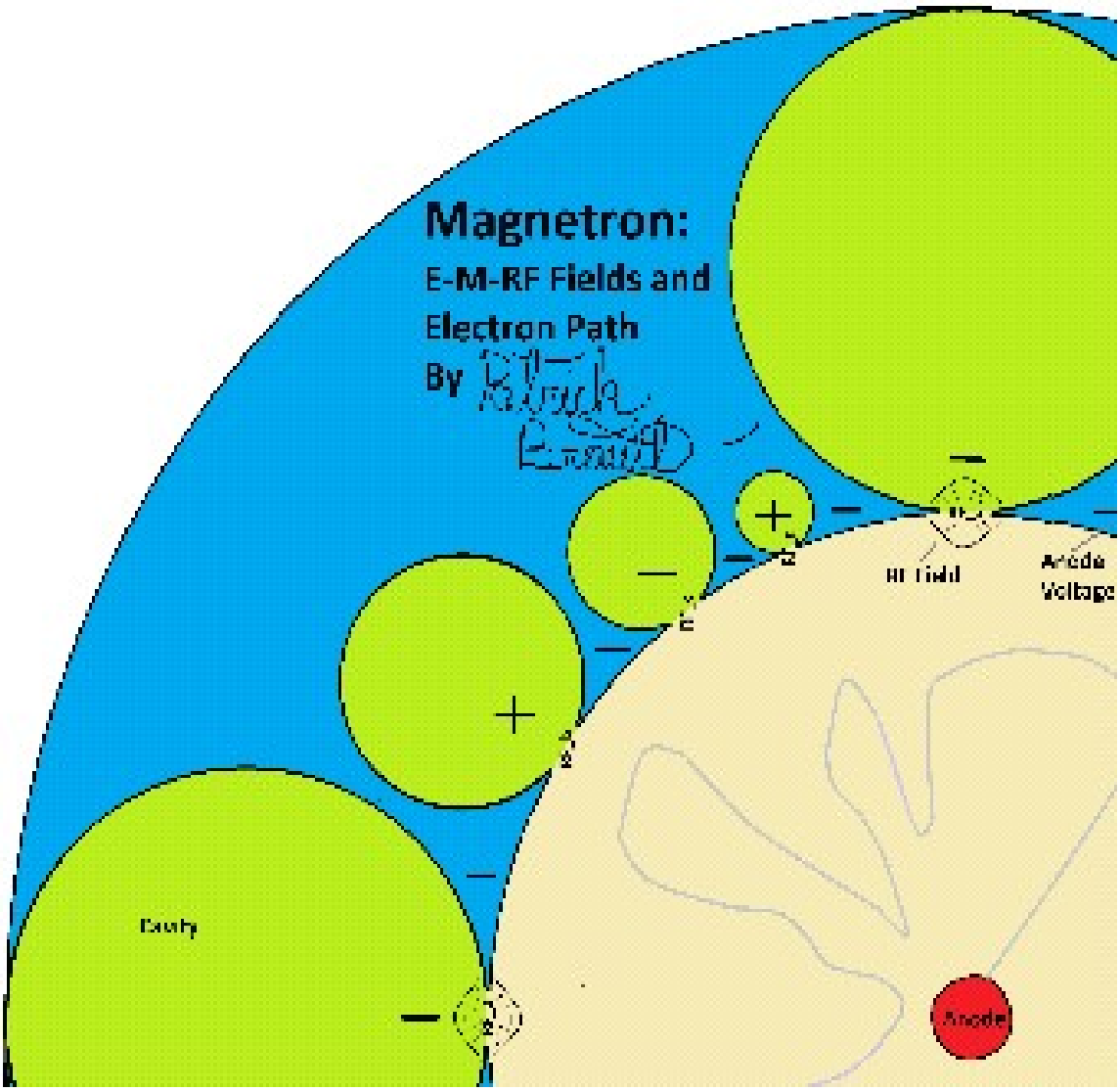
The following is an excerpt from Boots and Randall book that explains how an electron gains energy until it breaks the voltage barrier,

"In the magnetron an initial distribution of electric
field is assumed and orbits are calculated for this field. These
initial orbits enable the charge distribution to be found, giving
rise to a field of its own which, superimposed on the initially

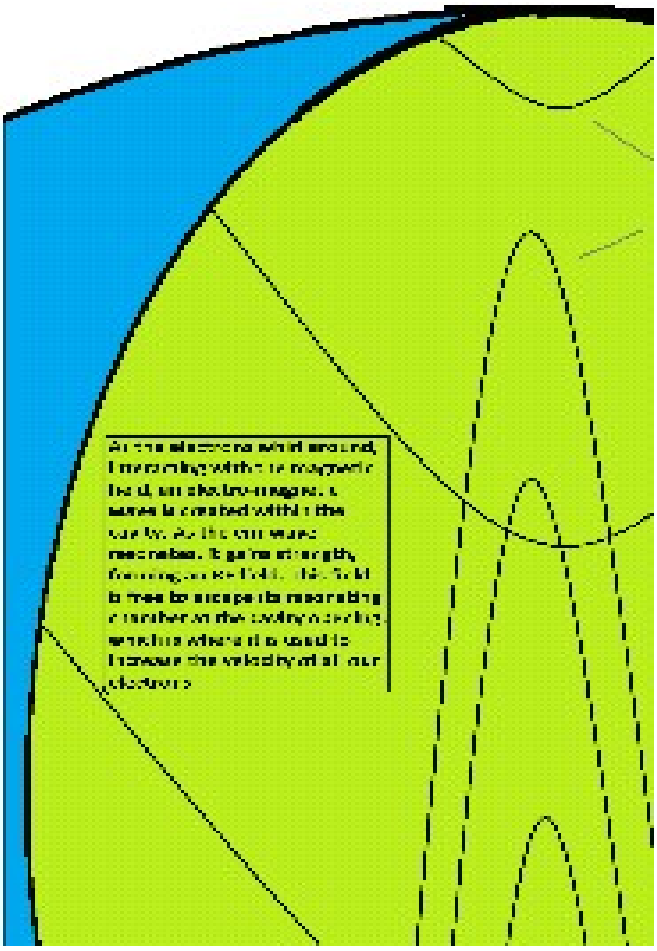
assumed field, gives the final field. This process is repeated with the corrected field until the initial and final fields agree sufficiently well. In Fig. 6 the electron distribution found in an eight-segment magnetron is shown at one particular instant. The square-shaped space charge is found to rotate in the direction indicated by the arrow and whilst the distribution does not change greatly, individual electrons move relative to the whole pattern. Each curve shown in Fig. 6 represents the position of electrons leaving different points on the cathode at a particular instant previous to that depicted by Fig. 6. The curve nearest the cathode represents electrons which were emitted one quarter of a period earlier. And consequently have not yet reached the region in which there is much effect from the oscillating segments. The curve showing maximum radial extension at the corners of the charge cloud is for electrons emitted two periods earlier and these have already reached the anode, as have those starting, periods earlier."



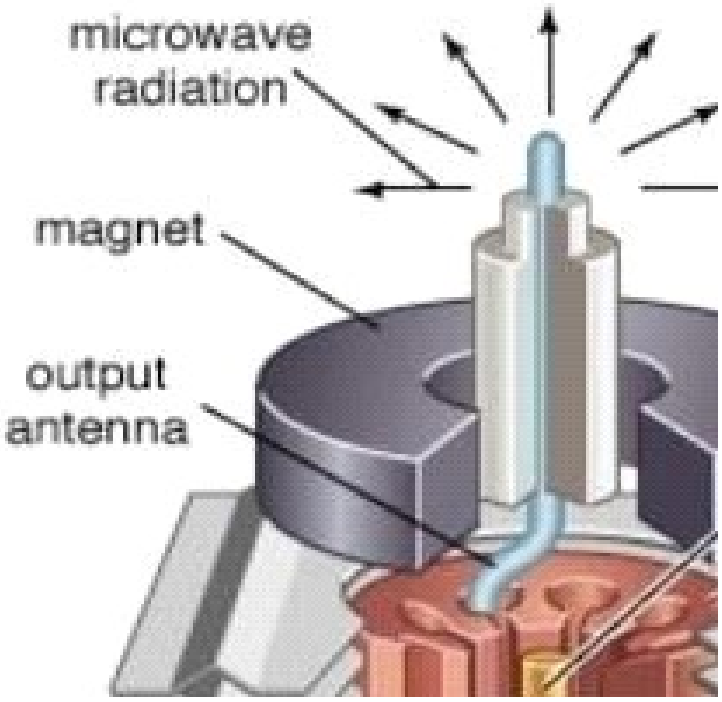
In the second image, below, I am showing a changing RF Field in agreement and opposition to the constant anode voltage. This is done to increase electron velocity so it can escape the chamber quicker and with more energy. The path shown is equivalent to the path of a f=-2 or f=-1 3/4 electron as explained in the above excerpt.



Next is an image that is an elongated version of the previous resonance chamber. This displays how an RF field is created. By a bunch of em-waves bouncing around. Except this image prompts the question, what if we change the shape of the resonance chamber. Obviously, this changes the RF field that is created.

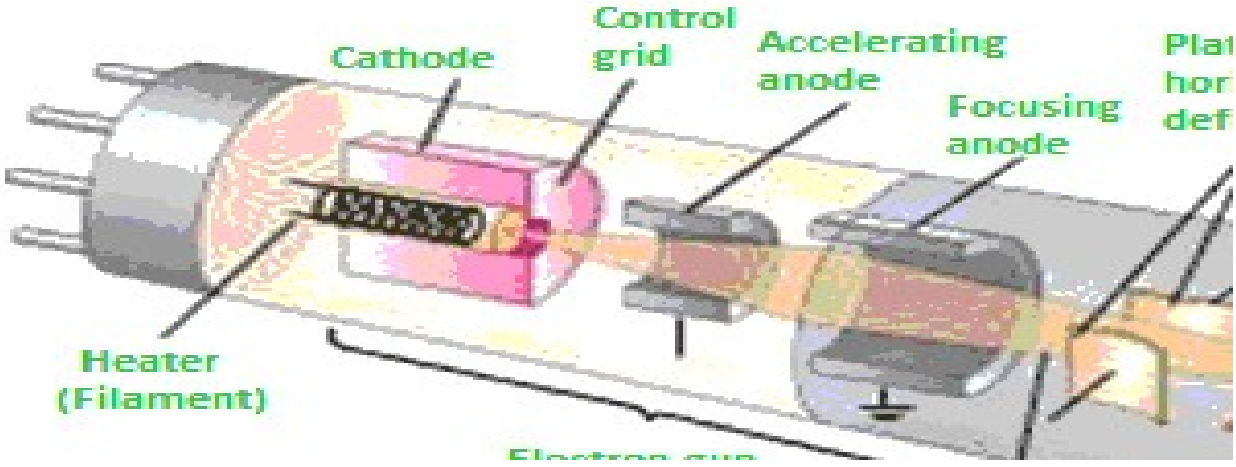


Finally, this next image displays the path of the electron as it leaves the main magnetron chamber, noted by the blue line. It is a clip from a previous image but I think it is important to see how a beam is created.



6.5.4 The Barrel

All the above images describe what is happening inside the magnetron. Together, they encompass the entire path of the electron, from beginning to end, going through the magnetron. Not shown, is the barrel and the various attachments that focus the beam. Consider the electron path after leaving the chamber but before leaving the barrel. The area I am referring to is just before the area labeled 'microwave radiation'. The barrel would look something like this:



What you are seeing in the above image are a series of anodes and magnets. They steer the em-radiation. Focusing it so it comes out the end of the barrel in a nice tight stream.

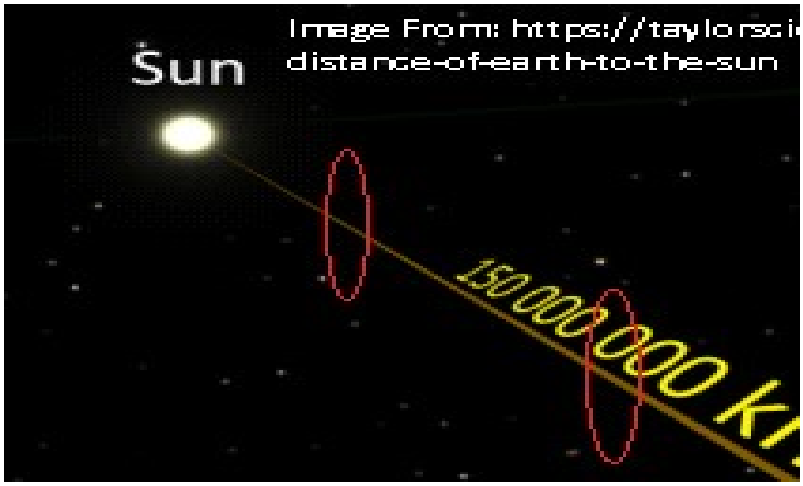
6.6 Anode Ring

Enter the Anode Ring. Like the barrel, an Anode Ring focus' and steers the energy beam. The electron beam never touches any of this equipment. It simply whizzes by, having its path changed a little. An Anode Ring is an anode in the shape of a ring, and it can be as large as we want it to be. So, to keep our beam from losing too much focus, we send it through a series of Anode Rings. Maybe relay station is a better word. Which I will talk about now.

6.6.1 Beam Travel

As our beam flies through space, on its way to Earth, the beam becomes spread out. You can think of it in terms of how a beam of light comes out of a flashlight, how the light becomes less focused. There may be times where, not only does our beam of electrons need to be more focused, but also its direction changed. This can be done with an Anode Ring, or a series of them. Similar to how a cathode and anode interact inside a cathode ray tube (crt) shown above. Where, electrons being shot out of the crt barrel are attracted to the anode, kind of like two magnets are attracted. Except, in this case, the anode is shaped like a ring, not a pole. So, the electron never hits anything directly. Instead, they whizz by, very close. Kind of like how a satellite uses a planet's gravity for Gravity Assist to help in braking and accelerating. The electrons come close enough to use the anode's electromagnetic field to steer.

Anode Rings, acting like relay stations, will be placed, above the ecliptic plane, in segmented locations between Earth and Sun. Picture a huge ring, floating in space. Similar to how the Anode in the barrel works, a number of Anode Rings will be placed along a path, each used to steer and focus the beam in a specific way.



The above image shows the Anode Rings, in red, placed in between the Sun and Earth.

6.7 Free Space Path Loss

Now to discuss free-space path loss (FSPL). This is the loss in signal strength of our electromagnetic wave. It is classified as the result from a line-of-sight path through free space, with no obstacles nearby to cause reflection or diffraction. Which is what you see in the image above except for one small detail. We will not be interested in the whole path from Earth to Sun. Only the path in between Anode Rings.

Free-space power loss is proportional to the square of the distance between our Anode Rings. It is also proportional to the square of the frequency of the radio signal.

$S = Pt / (4 * pi * r^2)$

The spreading out of electromagnetic energy in free space is determined by the inverse square law, which is displayed here.

Where, Pt = transmitted power [Watts], S = is the power per unit area at distance, r = Distance transmitter - receiver [Meters]

The receiving antenna's aperture, which describes how well it picks up power from the incoming electromagnetic wave. For an isotropic antenna, this is given by

$Pr = S * \lambda^2 / (4 * pi)$

Where, Pr = received power [Watts], S = non-directional power density, λ = transmitted wave length [Meters]

The total loss is given by the ratio:

$FSPL = Pt / Pr = (4 * pi * r / \lambda)^2 = (4 * pi * r * fr / c)^2$

Where, fr = transmitted frequency c = the speed of light

Keep in mind that, the shape of the antenna will be different at different points along the electron beam's path. This equation is for an isotropic antenna, one that receives a signal from every direction. The actual shape of the antennae will more than likely need to be a dish receiving antenna.

6.8 Earth Systems

The main purpose here is to catch the power being carried by the em wave and convert it into direct current, which can be used in the terrestrial power grid. You can think of this as being similar to catching a baseball with a mitt. Our microwave dish antennae, is the mitt and the beam is the ball. The Catcher is Earth. And safely protected, I might add. We can expand this concept a little further by placing a station in orbit and another on Earth's surface. Let's call them Orbital Transmitting Station and Earth Receiving Station.

6.8.1 Transmitting Station

Concerning the Orbital Transmitting Station, I will be referring to a review of technology that was recently written. It can be viewed, here. In the review, it is stated that Radio Frequency Energy Harvesting (RF-EH), is wireless electricity. That it can be used to transport energy through our atmosphere to various ground based stations. See image below for what this might look like.



This may look a little dangerous, and there are dangers involved, which were described in the earlier section on ionizing waves. This energy can be beamed to a remote location and inserted into our grid with no danger to human life. Radio waves carry an oscillating signal, just like our em-wave. They are less energetic and therefore, technically, less harmful. This behavior is mathematically and mechanically explained in such detail and quality in the above mentioned article that I will attempt to outline my own narrative using the information provided in it as a basis. Note that this article is simply a review of this technology. The narrative I wish to propose is a node based power grid that is fed by our gigantic em-wav coming from the Sun. This grid surrounds Earth, consisting of only a few orbital facilities that feed various ground based facilities. No more than a few per hemisphere, plus backups.

6.8.2 Receiving Station

The Earth Receiving Station is simply a microwave dish antenna that is placed on the surface of the Earth. Like the one shown below.



The dish antenna plays the role of the catcher. Moreover, this station will convert our wave into direct current and feed it into the power grid.

The size of the receiving dish antennae will be determined by the size of the beam. Specifically, the cross sectional area of the beam. To understand this, lets discuss the effective aperture and see how it interacts with the microwave beam. The beam hits the Reflector and is reflected toward the Receiver. The effective aperture, in this case, is located here, outlined in red in the image below. The effective area or receiving cross-section of the dish needs to be the same size as the cross-sectional area of the microwave beam in order to capture all its energy.



This is how the efficiency of an antenna capturing the power of incident electromagnetic radiation is measured. It is not necessarily the same as the antenna's physical area.

6.8.3 The Incoming Signal

Now to discuss how direct current is made from all this. There are some terms that need to be introduced. I will do that now. First, I will be relying on the theory of electrical and magnetic impedance of free space. Impedance just means, to be resisted by something. This is also known as the characteristic impedance of a vacuum. In terms of waves, this is the wave impedance of a plane wave as it travels in free space. You can think of it as how electro-magnetic waves naturally flow through free space. When the beam is generated, at the magnetron, it is the most concentrated. As it travels through space, it meets resistance, or impedance if you prefer. This resistance is not the same as that from rain or clouds, as described earlier in the section on atmospheric interference. This is the natural way the electro-waves interact with the micro-waves as they fly through space together. Both waves oscillate, with means their values change from negative to positive for a time and then back again. Each wave does this. However, when they travel together, they form a single wave. One wave that oscillates in two different waves. This is the Microwave. Just like a radio wave. This will be important later when discussing how alternating current is directly created from this signal.

The next two terms I want to introduce are Electric Field Strength, or Potential Gradient, and Current Density, or Magnetic Field Strength. I will use Electric Field Strength (EFS) and Current Density (CD). These terms are important because, at this point in the game, the voltage and current are spread out over an area, called the cross sectional area of the beam. Because the interaction between the wave beam and dish antennae require this distance parameter, the voltage becomes voltage divided by area. The terms voltage and current, as it refers to direct current will be discussed shortly. First, some equations.

This is the total power output of our beam.

Electrical Power (Pe) = 4E^15 Watts

At even one-percent efficiency that's, 4E^13 Watts.

The accepted value for the length of a microwave.

Wavelength (λ)= 10^-2 Meters

The accepted value for impedance of the beam in free space.

E / I = 377 ohms

The cross-sectional area of the beam (Ab) will be determined by the anode rings and magnetron, for now I will use an arbitrary value.

Using this and the equation P = I * E. A series of equations can be used to find values for the Electric Field Strength. Which was voltage divided by area and Current Density which was current divided by area. Remember, that P is equal to 4E^15 Watts.

For P = I * E and E / I = 377:

E =

I =

This is the total voltage and current that is spread out over the effective aperture of our dish antennae. Over the cross sectional area of the beam.

Next, I will discuss what is going on inside our Receiver and how this electricity ties into the power grid. The size of the effective aperture is found using a simple equation. However, this will be determined by characteristics of our wave beam. An antenna will have to be built to meet these required specifications. Using the following equation,

$A_e = \lambda^2 / (4\pi) * G$

Where A_e is effective aperture, λ is wavelength, and G is antennae's gain.

Unlike a wire that carries a current of electrons, the flow of electricity, microwaves themselves don't directly carry electrons or transport electricity. For a microwave, the beam's cross sectional area interacts with the metal in the antenna, the oscillating electric field of the microwaves pushes and pulls on the free electrons that are in the antenna, creating an alternating current.

6.8.4 Direct Current Electricity

A rectifying antenna can convert the microwave beam into direct current. Which is simply the addition of another circuit to the dish antenna. This circuit, a rectifier circuit, is an electronic circuit that converts alternating current into direct current. It uses diodes that only allow current to flow in one direction, acting as a one-way valve for electricity.

Chapter 7: Assembly and Material Logistics

7.1 Primary Transport

-Achieved velocity at 28,000 kilometers per hour.
-Would take approximately 5321 hours or 222 days to travel from Earth to the Sun.
-Payload capacity of 100–150 tons to low earth orbit and 27 tons to geostationary transfer orbit.
-Since they are the best for the job, I am using the same ship and rockets, Starship with Raptor Rockets, as I did in my Master's Thesis. For now, I am using the same numbers too. These numbers will get us about a third of the way to the sun. The first round of numbers I need to use for fuel calculations are basic stats about Starship. Other numbers were determined by smart people at Nasa for missions to Mars. Unfortunately, I do not have data specifically for Starship or the Raptor Engines for a trip to the Sun. So I will be making assumptions on the available data that I do have.

I will use a dry mass(Dm) of 120000 kg for Starship. A Max Payload capacity (Mp) of 100000 kg. Its max propellant (Pm) capacity is 1200000 kg. Raptors Isp in a vacuum (Isp) is 380 seconds. The oxidizer to fuel ratio (Ra) is 3.55. I will start with a change in velocity (deltaV) of 6000 m/s. Lowering the deltaV means a longer trip time with less fuel costs. Which is ok, since these starting values were meant for human cargo, not equipment. The gravitational acceleration (g) I am using is 9.81 Meters per Second Squared. From this info, we can determine the initial mass (Im) and then the total propellant(Tp) required to travel from Low Earth Orbit to the Sun, which we will call the Transit to Sun burn.

Im = Dm + Mp + Pm = 1420000 Kilograms

Tp = Im * (1 – e^(-deltaV/(isp * g))) = 1420000* (1-e^(-6000/(380*9.81))) = 1136181 Kilograms

So, to move 100 tons of equipment from Low Earth orbit To the Sun would take about1136181 Kilograms of fuel. But wait! That is only one third the distance. However, if we change the deltaV to a smaller number, then the same amount of fuel will take us farther. Now let's try it with a deltaV of 1000:

Tp = Im * (1 – e^(-deltaV/(isp * g))) = 1420000* (1-e^(-1000/(380*9.81))) = 334108 Kilograms

Now, since Starship's fuel capacity is 1200000 kg, and if we use a deltaV of 1000, we should have approximately enough fuel to get us to the Sun carrying 100 tons of cargo each trip.

So how many trips is this going to take? That is easy. For a total weight of 1.8E^11 Kilograms. It would take 1.8E^11 / 100000 = 1.8 million trips to move all this material to the Sun. Sounds like a lot right. Remember, this is all calculated on assumptions, and no matter how reasonable an assumption you can make, even a little deviation will cause a big change in the total number of trips it takes to move all this stuff. So why

am I calculating this using shady numbers? Because, this is just a starting point. These are starting values. For example, the number 1.8 million is if we are only using one ship. If we use two ships, that number divides in half, and so on. Once everything is up and running, I think it is a safe assumption that all the material for the solar sail could be moved to the 20 yd line in 100.6 years. This is with 14 ships, not one or two. Which is within reason, for this point in the game.

7.2 Material Weight

-The total weight of our solar sail was calculated to be 1.8E^14 Kilograms.
-There is a difference of 1.4E^13(pull) - 6.8E^10(push) = 1.4E^13 Newtons between the pull of the Sun's gravity and the push of its radiation on our solar sail alone. Surely, increasing the area of our solar sheet will be required since we will be adding even more weight from the Magnetrons, Capacitors, and circuitry that is needed.
-The weight of the rest of the equipment is going to be difficult to estimate. So, for now, let's just calculate the weight of the solar sail, without the magnetron, capacitor or circuitry.

Chapter 8: Lifespan Maintenance

8.1 General Maintenance

-Solar sails will be the only element that needs regular maintenance. This is ok, as heritage grows, as solar cell efficiency increases, as the entire system enters a state of equilibrium, the solar sails will need to be replaced. All other equipment, like the magnetrons and capacitors, sit behind the solar sail and are protected. This equipment requires little maintenance.

8.2 Solar Sail

-Short Lifespan.
-Solar sails will most likely be damaged by various emissions from the Sun.
-This is alright, replacing solar sail components will be required as system heritage grows
and photovoltaic cell efficiency increases.

8.3 Other Components

-Anode Rings, Earth Orbital Receiving Station, Ground Station,

8.3.1 Capacitor

8.3.2 Magnetron

-Long lifespan.

-These components are durable and do not wear easily.

-They are relatively safe from any Sun's emissions and there is plenty of heritage in them operation.

Chapter 9: Efficiency

Efficiency will only get better. For many reasons. There is a lot of room for improvement in the solar panels. Also, at this point in the game, it is very difficult to say how efficient a large capacitor or magnetron will be. One thing is for sure, it will take many trial versions to get the system into equilibrium, once that is done, then we can get exact answers for efficiency. So, for now, I am going to use my best guess to calculate an example of usable wattage that is actually injected into the power grid.

Solar Panels: 20 percent efficient

Solar Storm (sail to capacitor): 90 percent efficient

Capacitor: 90 percent efficient

Magnetron: 80 percent efficient

Transmitting station: 50 efficient

Receiving Station: 80 percent efficient

Ac/Dc Conversion: 70 percent efficient

Remember that the full scale version of the solar sail produces 42 quadrillion watts. Inefficiencies are encountered at each component. Here it is:

42 * .2 * .9 * .9 * .8 * .5 * .8 * .7 = 1.5 quadrillion Watts

Which is 1.5 trillion kilowatt-hours. So, as you can see, even after efficiencies are taken into account the amount of wattage is astounding.

Chapter 10: First Steps

First step is to get material for a small scale operation. To begin coordinating resources. To get your foot in the door, so to speak, by having the discussion and making plans. Explore what it would be like to have a constant stream of cargo headed to the sun. Practice by setting up a sample staging area for incoming material, an assembly area for work in progress, and a final waiting area. Venus, or its orbit rather, is a fine location to setup operations for these first steps. However, the equipment will ultimately need to get closer to the Sun.

Assembling large sections of the solar sail at such a Mercury base and moving them into location above the solar poles is probably the easiest way to build a final assembly. The difficult part will be finding a location where the sail floats between solar radiation and gravity. To do this, a solar sail may need to be setup and allowed to settle into place. From that location, or distance to the Sun, more sail sections can be added. Until, eventually, the surface area of the sail is large enough, creating enough push from solar radiation, to support equipment, like the capacitor and magnetron, against the pull of gravity.

The first step is all about building heritage. The entire operation, everything, all the components, the system as a whole, only works at time equals infinity. When the system is in equilibrium. Which is only when all its components parts are in their own running equilibrium. Therefore, building system heritage by setting up a working model is the first step.

Chapter 11: What About the Sun

The Sun's lifespan or solar cycle will not be affected. Once at full scale, there may be a tiny rise in temperature at the poles, under the solar sheets but it would be a micro-small increase.

Chapter 12: Theoretical Application of EM-Wave Technology

Sections one and two are specific to the Magnetron. Section three is an extension of this equipment into the quantum realm and is outside the scope of this paper. However, section three, is based in Newtonian Mechanics and EM-Wave technology.

12.1 Anti-Gravity

Just what it sounds like. Flooding an area with this type of em-wave will result in matter becoming weightless.

12.2 Phase Shifting

This is going to be a difficult concept to grasp for most people. Using elements of quantum mechanics, like quantum spin, to explain this makes sense but only if you understand quantum spin. For the rest, this can be compared to polarized sunglasses. Just as your polarized sunglasses block out, and let through, only specific wavelengths of light. An em-wave can cause matter, like a wall or solid stone, to have a quantum spin that is dissimilar to our own.

12.3 Sponteneous Down Conversion and Viewing the Possible Futures

For this section, I will be referring to the work that has recently been done in Spontaneous Parametric Down Conversion to explain how it is possible to view an array of outcomes for a specific time or event. Similar

to what one might see if they had the ability to flip through the multiverse and view different outcomes of one event. And when I say view, I mean just as if you were watching them on a tv or computer screen. This section depends highly on the existence of dimensions beyond the four that we know. Specifically, it needs the current theories of the multiverse and string theory to be true. Currently they are just theory. However, electromagnetic equipment like the cathode ray tube can manipulate particles on a screen. This is exactly how the first televisions worked. By quantum entangling the phosphorus crystals on the back side of your old television screen, making this now a quantum system, obeying the rules of the quantum world. Now the em-wave being emitted from the cathode ray tube makes a projection the screen but is doing it in a quantum system, one that obeys the rules of quantum mechanics. Not the rules of the Newtonian world that we are all too familiar with. In the quantum world, the entangled crystalline structure of the phosphorus allows for use of the rules of the quantum realm. So by changing the em-wave that comes out of the cathode ray tube, you would inherently be making changes to a quantum system. The same quantum system that is responsible for making the images on the screen. Images that come from a different series of events with the multiverse that we all live in.

12.4 How It All Works

In the activities mentioned in the above two sections, a specialized cavity magnetron creates a field from a direct current source. Inside the magnetrons main chamber, the configuration of the magnetron determines the properties of the field that is created. As mentioned before, the configuration of the magnetron includes many aspects on the inner workings of the main chamber. This includes how the rf fields are created, from the resonating chambers. Also, the design and layout of the teeth and the strength of the magnetic and voltage fields. This is a different configuration than the one that is used in the cavity magnetron that is mentioned in chapter 6.5. One main difference, is that there is no barrel or mechanics that focus the magnetrons output to create a beam. This instead, is a broad field em-wave.

12.5 Applications

Combining two magnetrons, one of antigravity configuration and the other of phase shifting configuration is the main topic of this section. For example, aiming these two fields at the same spot on the ceiling of a two story home would allow you to jump up through the ceiling into the second floor. Really, think about it. The anti-gravity field allows you to jump very high and the phase shift field allows you to go through the ceiling. So, running and jumping, in the correct fashion, could literally land you on the second floor somewhere. Taking this simple concept and applying to this the main topic of this paper. It can easily be seen

how lifting all the required material out of Earth's orbit would benefit from using this technology in this fashion.

Definition List:

Variable List:

Name:	Abv:	Value:	Unit:
# of Homes Powered Solarsheet Sample Earth	Hes	Pes / Wavg	
		11128068183	Homes
# of Homes Powered Solarsheet Sample 20 Yd Ln	Hys	Pys / Wavg	
		2.78E+11	Homes
# of Homes Powered Solarsheet Full 20 Yd Ln	Hyf	Pyf / Wavg	
		3.39E+13	Homes
Power from Solarsheet Sample at Earth	Pes	Ie * As	
		1.36E+13	Watts (W)
Power from Solarsheet Sample at 20 Yd Ln	Pys	Iy * As	
		3.41E+14	Watts (W)
Power from Solarsheet Full at 20 Yd Ln	Pyf	Iy * Af	
		4.15E+16	Watts (W)
Power from Solarsheet Full at 20 Yd Ln	Pyf	Pyf / 1000	
		4.15E+13	KiloWatt Hours (kWh)
Solarsheet Sample Size Area	As	1.00E+10	Meters Squared (m^2)
Sun Surface Area	Sa	6.10E+12	Meters Squared (m^2)
Solarsheet Full Size Area	Af	1.22E+12	Meters Squared (m^2)
Distance Earth to Sun	De	1.49E+11	Meters (m)
Distance 20 Yd Ln to Sun	Dy	2.98E+10	Meters (m)
Solar Irradiance 20 Yd Line	Iy	Savg / (4 * pi * Dy^2)	Watts per Meter Sqr
		3.41E+04	(W/M^2)
Solar Irradiance Earth	Ie	Savg / (4 * pi * De^2)	Watts per Meter Sqr
		1.36E+03	(W/M^2)
Avg Watts Per USA Home	Wavg	1224	Watts (W)
Avg Solar Power	Savg	3.80E+26	Watts (W)
Market Panel Weight	Sm	2	Kilograms (kg)
Market Panel Area	Ss	1	Meters Squared (m^2)
Advanced Panel Weight	Ssm	0.15	Kilograms (kg)
Advanced Panel Area	Sss	1	Meters Squared (m^2)
Gravity Equation	F	G * (M1 * M2) / D^2	N*m^2/kg^2
Gravity Constant	G	6.7E-11	N*m^2/kg^2
Mass of Sun	M1	2.00E+30	Kilograms (kg)